

APPENDIX · TECHNICAL IMPLEMENTATION

Delivery format?	A web app with phone and desktop layouts from the same page, any browser. Two windows on stage stand in for two phones. Native mobile is roadmap.
Technologies?	Node/Express API, Redis Streams event log, Postgres read models, vanilla JS + WebCrypto AES-GCM, nginx, Docker Compose. Node's built-in test runner. Vended QR library. No frameworks, no CDNs, so the demo needs no internet.
Behind the scenes?	Phone encrypts → POST /events → membership check → XADD to the stream → projector consumer group → Postgres events + handoffs + workload view, then an announcement on a Redis channel that the API relays to every open tab as a server-sent event; a second consumer group, the notifier, writes alerts per subscriber → phones poll and decrypt. The dashboard is computed on the phone.
Built from scratch here?	Spec, requirements and two audit rounds were planned ahead. All code was written during the hackathon window. QR generation is a vended library.
Tradeoffs?	Server-sent events over mobile push, a demo default password over real credential policy, one stream over per-family streams, consent filter deferred. Each one on the honesty slide with the reason.

APPENDIX · IMPACT AND USERS

Target user?	Adult children sharing care of an elderly parent. The elder is the subject and owner of the preferences, not the app user. Support workers join with their own role.
Discovery and adoption?	Family to family, by QR, in person, by design. Community care programs and masjid welfare committees as the channel. Agencies bring their families.
How many benefit?	1 in 4 American Muslims already care for an older adult (ISPU 2026). Muslims are 8.7% of Greater Montreal (Census 2021). Start with Montreal families, then any family with a shared care load.
Impact if fully developed?	Fewer missed meds and appointments, a fairer split of the load, and less of the strain ISPU documents. A family-owned health record the family never had.
Success measure?	Plan adherence, handoff acceptance time, active caregivers per family, weekly return rate.

APPENDIX · DEMO, FUNCTIONALITY, BUSINESS

Fully functional or mocked?	The core loop is real end to end: encrypt, stream, project, store, decrypt. Login is real, with a demo default password. Updates are live over server-sent events, with polling as the safety net. The consent filter is not built. Nothing on stage is faked.
Completed vs prototyped?	Completed and running: family create and join, login across several families, the routine, today's plan, logging with undo, the handoff, alerts, the elder screen, the hospital sheet, recommendations, Server view, 119 tests. Prototyped: nothing on stage. Not built and labelled so: the elder consent filter, key rotation, push, Urdu and Arabic.
Most important feature?	The record itself, private by construction. From it: the handoff that writes itself, and the hospital sheet the ER can read in a minute.
Hardest part?	Keeping the server useful while it can read nothing: deciding which fields stay clear for routing and counting, and building a whole dashboard from client-side decryption.
Business model?	Free for families, always, with a voluntary support option for extras only. Paid by agencies and care organisations at \$300 per elder a year on one invoice, and by support contracts for self-hosting. Community organisations can instead sponsor the free Family tier on a yearly package, about \$600 for 50 families, which is not the professional licence. The server, about \$1,000 a month, is covered by 40 elders; \$5,000 a month by 200. Estimates, to validate with five pilot families and two agencies. Full model in pitch/BUSINESS.md.
Scale beyond the prototype?	Per-family streams, push, scoped keys, password policy, offline. Ciphertext-only storage keeps compliance and hosting light.
Who pays?	Agencies and care organisations per elder, community organisations per year for the families they sponsor, organisations that self-host for support. Families never pay for the record; some choose to support it.
What would it take to make it a real product?	Password policy and key recovery, native apps, scoped support keys, push, offline, accessibility, Urdu and Arabic.

APPENDIX · TEAM, PROCESS, DATA AND PRIVACY

How did you divide the work?	Abdul: architecture, event model, crypto, api and projector, dashboard, tests. safio: in-app invite flow, interface cleanup, the flow view that traces a write hop by hop.
What did you learn?	Cut scope early and say so. One workflow done well beats five half done. Privacy shapes every feature, including which ones you cannot build yet.
Differently next time?	Tests from hour one. Push notifications budgeted from the start.
How did you prioritise?	Requirement register of 46 items, two audit rounds on feasibility in 24 hours, then "cut core": the handoff loop first, everything else labelled roadmap.
What data?	Care events, handoff summaries, preferences, routine, member names, all encrypted. Clear: event type, category, member id, timestamps.
How do you handle privacy and security?	The family key is generated in the browser and never leaves it. The server stores ciphertext plus routing metadata, and its own database is unreadable without the key. Joining is a QR code handed over in person. Anyone can check it live: Server view shows any screen as the server sees it. Open limits: support workers hold the same key, the demo password is 333, and there is no key recovery yet.
Ethical risks?	Support workers currently hold the same key, so hiding categories is UI only. The password-locked key copy is only as strong as the password, and the demo default is 333. Lost key means lost data until rotation ships. No diagnosis, dosage or triage anywhere, and the metrics are counts, never judgments.
Depends on an LLM?	No. By design the server cannot run one on this data.

APPENDIX · FUTURE AND CLOSING

Next if more time?	Urdu and Arabic for the elder view, scoped support key, push, key rotation, password policy and recovery.
Continue after today?	Yes. Five-family pilot in Montreal next week.
First step after today?	Commit and publish the repo, deploy on one small VM, recruit the pilot families through community care contacts.
Different from existing options?	Caregivers on Reddit name the same tools over and over: shared Google Calendar, iPhone Notes, Cozi, CaringBridge, Caring Village, Lotsa Helping Hands, FamCal, a dry erase board, a paper binder, blister packs. Every one is a calendar, a message thread, a reminder or a pill device, and every one is readable by its vendor. None is the record of what was actually done, by whom, against the family's own plan, with the handoff and the load in the same place.
Help needed?	Introductions to community care programs and agencies, pilot families, and a designer for the elder view.
Remember most?	The server cannot read it. Tap Server view on any screen, or open the database, and see for yourself.
Why should this win?	It solves the coordination problem the track asked for, with the elder's preferences built in, privacy that is cryptographic rather than a promise, an honest scope, and a test suite behind it.

APPENDIX · OBJECTIONS WE EXPECT

An app is not the problem, willingness is	Put to an engineer offering to build exactly this, on r/CaregiverSupport, 20 Oct 2023. Fair. The app does not create willingness. It removes the excuse of not knowing, and it makes an unequal load visible so willingness has something to act on. Where a family is genuinely unwilling, nothing we ship fixes that.
Tracking becomes surveillance	Real risk. On r/dementia a woman is required by her brother to log her mother’s waking, meals, moods and bathroom breaks daily, and the replies call it control. Our record is symmetric: everyone sees the same thing, including the workload view, so it cannot be one sibling auditing another. Counts only, never judgments, no diagnosis or dosage.
Nobody wants another app	Also real, and stated on r/AgingParents. That is why joining is a QR code handed over in person by someone the family already trusts, why families never pay, and why the elder is never asked to enter anything.
One sibling will refuse to use it	Reported repeatedly about shared calendars. The product stays useful with one person logging: the record, the hospital sheet and the handoff all work single-handed, and the workload view then shows the imbalance in numbers instead of in an argument.
Why trust your encryption?	Do not trust it, check it. Server view shows any screen as the server sees it, and the database rows are on the backup slide. A privacy test in the suite asserts that a logged note appears in no server row, no stream entry and no API response.
What breaks first?	The key. Today there is no rotation and no recovery, so a family that loses every device loses the record. That is the one assumption we have not de-risked, and key recovery is the first thing built after Sunday.

APPENDIX · SOURCES

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